PLANT LEAF DISEASE DETECTION USING ARTIFICIAL INTELLIGENCE

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PROJECT PROPOSAL

on

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1 Introduction

Agriculture has always been the backbone of the Indian economy, and the health of crops plays a major role in deciding the overall productivity of farmers. In recent years, the importance of early and accurate detection of plant diseases has increased because even a small infection on the leaves can spread rapidly and damage the entire crop. Traditionally, farmers depend on their personal experience or advice from local experts to identify diseases. However, this approach is slow, subjective, and sometimes inaccurate, especially when the symptoms of different diseases look similar.

With the advancement of Artificial Intelligence (AI) and Computer Vision, the process of diagnosing plant diseases has become faster and more reliable. By using images of plant leaves and applying deep learning models, it is now possible to detect diseases automatically with high accuracy. This method reduces dependency on manual observation and helps farmers take preventive measures at the right time. Such technology is extremely useful in a country like India, where a large portion of farming still happens in rural areas with limited access to expert-level plant pathologists.

Plant leaf disease detection is not just a technical concept; it is a real-world solution to a major agricultural challenge. Many deep learning architectures like Convolutional Neural Networks (CNN), MobileNetV2, and ResNet are widely used for image classification tasks. These models can learn the visual patterns of different plant diseases from datasets and recognize them in new images. However, certain issues such as variations in leaf color, uneven lighting, complex backgrounds, or very small disease spots make the task challenging.

Recently, Vision Transformer (ViT) based models and self-supervised learning approaches have shown excellent performance in image understanding problems. Inspired by these improvements, our project explores a DINOv3-based ViT approach for plant disease detection. The aim is to understand how traditional CNN models compare with modern transformer-based architectures and which architecture performs better for real-world agricultural applications.

In this project, we present a complete workflow starting from understanding the need for plant disease identification to experimenting with multiple models and finally evaluating their performance. The motivation behind this work is to contribute towards affordable and efficient AI-based agricultural solutions that can help Indian farmers protect their crops and improve yield.

2 Related Work

Identification of plant diseases using leaf images has been an active research area for many years, especially after deep learning became popular in image classification tasks. Several researchers have applied different machine learning and deep learning methods to improve the accuracy and speed of disease detection. Earlier studies mainly relied on manual feature extraction techniques, where texture, color, and shape features were designed by experts. Methods such as Local Binary Patterns (LBP), Gray Level Co-occurrence Matrix (GLCM), and color histogram analysis were widely used to detect patterns on leaves. Although these traditional approaches gave reasonable results, they struggled when the dataset was large or when diseases had similar visual symptoms [1].

With the introduction of Convolutional Neural Networks (CNNs), the process of feature extraction became automatic. CNNs can learn important patterns such as spots, patches, edges, and color variations directly from the images without manual intervention. Many studies used simple CNN architectures for classifying diseases in crops like tomato, potato, rice, and maize. Researchers also experimented with deeper architectures to improve performance. For example, MobileNetV2 was used in several works because of its lightweight design and suitability for mobile devices. Its depthwise separable convolutions reduced computation, allowing faster detection in real-time systems [2].

Another line of research explored the use of ResNet models, which introduced residual learning to solve the problem of vanishing gradients during training. ResNet-50 and ResNet-101 were commonly used for leaf disease identification in large datasets such as PlantVillage. These models provided high accuracy and strong generalization ability but required more computational power compared to MobileNet-based approaches. Many comparative studies showed that ResNet models outperform simple CNNs but may not be ideal for deployment on low-resource devices such as smartphones used by farmers in rural areas [3].

Recently, transformer-based models have gained attention in agricultural image analysis. Vision Transformers (ViT) introduced the concept of patch embeddings and self-attention mechanisms to understand image features globally rather than locally. This is particularly helpful in plant disease detection where disease symptoms may appear in scattered or irregular patterns on the leaf surface. Self-supervised learning models such as DINO and DINOv2 further improved learning efficiency by reducing the need for large labeled datasets, which is beneficial in agriculture where collecting labeled images is difficult [4].

Several studies compared CNN architectures with transformer-based approaches and found that ViT variants often achieve better performance, especially in complex lighting or background conditions. Additionally, transformers can handle variations in disease pattern size more effectively. Research also shows that DINO-based models provide strong feature representations even without heavy training, making them suitable for real-world

agricultural applications [5].

Overall, past research indicates a clear shift from traditional image processing methods toward deep learning and transformer-based techniques. While CNNs remain effective and easy to deploy, transformer models such as ViT and DINOv3 are emerging as more powerful alternatives for handling diverse and challenging plant disease datasets.

3 Challenges and Issues

Plant leaf disease detection using AI is a growing research area, but it also comes with several practical challenges. These challenges occur because real plant images are captured in different environmental conditions, with variations in lighting, background, disease severity, and leaf structure. While deep learning models have improved accuracy, several problems still limit the performance of these systems in real-world agricultural settings [6]. The major issues can be grouped into three categories: image-related challenges, dataset-related challenges, and model-related challenges.

3.1 Image-Related Challenges

3.1.1 Illumination Variation

One of the biggest problems in plant disease detection is uneven lighting. Farmers take images at different times of the day, sometimes under very bright sunlight and sometimes in shaded areas. This causes changes in color intensity and contrast, which confuse the model. Deep learning models trained on ideal dataset images often fail when brightness levels change drastically [7]. Even modern CNNs struggle to differentiate between actual disease spots and reflections caused by sunlight.

3.1.2 Complex Background

In controlled datasets like PlantVillage, leaves are mostly captured against a plain background. But in real farm conditions, leaves appear along with soil, other leaves, insects, shadows, and agricultural tools. Complex backgrounds reduce model accuracy because the network may learn background patterns instead of disease features [8]. Segmenting the leaf from the background is also difficult when the leaf edges overlap with similar colored plants.

3.1.3 Disease Spot Variability

Different diseases show symptoms in different sizes and shapes. Some diseases create very small dots, while others produce large patches. Detecting early-stage infections is particularly difficult because the affected area may be too small for the model to notice. Moreover, two different diseases may show similar visual symptoms, which increases misclassification chances [9].

3.2 Dataset-Related Challenges

3.2.1 Class Imbalance Problems

Many datasets contain more images of common diseases and fewer images of rare ones. This creates a problem known as class imbalance. Models tend to overfit on dominant classes and ignore the under-represented ones. As a result, rare but harmful diseases remain undetected in real-life scenarios [10]. Data augmentation helps, but it cannot fully replace real image diversity.

3.2.2 Limited Real-World Data

Most open-source datasets contain laboratory-style images taken under controlled environments. These datasets do not represent the messy and unpredictable conditions found in farms. Collecting labeled field images is difficult because experts are required to correctly identify each disease. This lack of field data reduces the generalization ability of deep learning models [11].

3.3 Model-Related Challenges

3.3.1 Overfitting in CNN Models

CNN models such as MobileNetV2 and ResNet perform well during training but sometimes fail to produce similar accuracy on real images. This happens because the model memorizes the dataset instead of learning actual disease patterns. Overfitting is very common when the dataset is small or lacks variation [12]. Techniques like dropout and regularization help but do not completely eliminate the issue.

3.3.2 High Computational Requirements

Deep learning models, especially deeper CNNs and transformer-based architectures, require powerful GPUs for training. For many farmers and agricultural centers, such infrastructure is not affordable. Even though lightweight models like MobileNetV2 exist, their accuracy decreases in complex conditions. On the other hand, ViT and DINO-based models need more memory and computation [13].

3.3.3 Sensitivity to Noise and Occlusion

Leaves in farms are often partially covered by other leaves, insects, dust particles, or raindrops. This creates occlusion, making it difficult for models to detect disease. Noise in images due to camera shake or dust can also mislead the classifier. This is a serious challenge because farmers rarely take perfect images in the field [14].

Overall, these challenges show that plant leaf disease detection requires robust AI models that can handle diverse environmental and image conditions. Modern transformer-based models like ViT and DINOv3 aim to solve many of these problems by focusing on global image features, but further improvements are still needed for large-scale deployment in Indian agriculture [15].

4 Objectives

4.1 Aim of the Project

The main aim of this project is to develop an intelligent system that can automatically identify plant leaf diseases using Artificial Intelligence and Deep Learning techniques. The objective is to design a model that is accurate, reliable, and useful for real agricultural conditions, so that farmers can quickly detect infections and take preventive actions. The project also aims to compare traditional CNN-based models with transformer-based approaches like Vision Transformers and DINOv3 to understand which architecture performs better for plant disease detection.

4.2 Scope of the Project

The scope of this project includes the collection and analysis of plant leaf images, preprocessing of data, training and evaluation of different deep learning models, and comparison of their performances. The system is expected to recognize multiple disease classes and work effectively even when environmental conditions such as lighting or background vary. The project also focuses on creating a model that can be integrated into a mobile or computer-based application for practical use by farmers. By testing multiple architectures, the study aims to provide insights into building better and more adaptable plant disease detection systems suited for Indian agricultural environments.

5 Methodology

The methodology used in this project focuses on developing a comparative deep learning framework for plant leaf disease detection. Several models such as CNN, MobileNetV2, ResNet, and YOLO were implemented and evaluated. Finally, a DINOv3-inspired Vision Transformer model was explored due to its strong performance in learning global image features. This section explains the overall workflow, the models used, the dataset characteristics, and the evaluation techniques adopted in the study.

5.1 System Workflow and Architecture

The overall system for plant disease detection follows a structured pipeline similar to many deep learning-based vision systems. The workflow includes image acquisition, preprocessing, model selection, training, and prediction. A conceptual block diagram of the methodology would include the following steps:

- **Image Acquisition:** Leaf images are collected from publicly available datasets and manually captured images.
- **Data Preprocessing:** Images are resized, normalized, augmented, and prepared for training.
- **Feature Extraction and Training:** Different models extract visual features and learn disease-specific patterns.
- **Model Comparison:** CNN, MobileNetV2, ResNet, YOLO, and DINOv3-ViT are compared.
- **Prediction and Classification:** The best-performing model predicts the disease category for new images.

Block diagrams in simple reports generally visualize the flow of data from input to output. The structure used here is inspired by similar deep learning-based agricultural systems described in previous research [6, 8].

5.2 Preprocessing and Data Preparation

Before training, the images undergo several preprocessing steps. These steps ensure that the deep learning models receive clean, standardized input:

- **Image Resizing:** All images are resized to 224×224 or 256×256 based on model requirements.

- **Normalization:** Pixel values are normalized to bring the images into a common intensity scale.
- **Data Augmentation:** Techniques such as rotation, flipping, zooming, and brightness adjustment are applied to increase dataset diversity [11].
- **Label Encoding:** Each class label is encoded into integers for supervised learning.

These steps help the model generalize better and reduce overfitting, especially when class imbalance exists.

5.3 Model Descriptions

5.3.1 Convolutional Neural Network (CNN)

CNNs are widely used for plant disease classification due to their simplicity and effectiveness. A basic CNN model consists of convolution layers, pooling layers, and dense layers. CNNs extract local patterns such as spots, color patches, and edges, which makes them suitable for early disease detection [12]. However, CNNs struggle when disease patterns appear in irregular positions or when the background is complex.

5.3.2 MobileNetV2

MobileNetV2 is a lightweight deep learning architecture designed for mobile devices. It uses depthwise separable convolutions, reducing the number of parameters significantly. Advantages:

- Fast training and inference
- Suitable for smartphones used by farmers

Limitations:

- Lower accuracy in complex environments
- Sensitive to lighting changes and tiny disease spots

MobileNetV2 performed moderately well but was not robust enough for real-world field images [2].

5.3.3 ResNet

ResNet introduces residual connections that help solve the vanishing gradient problem. ResNet-50 and ResNet-101 were tested in this project. Advantages:

- Strong generalization ability
- Good performance on medium and large datasets

Limitations:

- High computational cost
- Slow inference on normal devices

ResNet provided better accuracy than CNN and MobileNetV2 but required more GPU memory [3].

5.3.4 YOLO for Real-Time Detection

YOLO (You Only Look Once) is a popular object detection model. Although it is mainly used for bounding-box detection, it was tested here to evaluate how it performs in identifying diseased regions. Strength:

- Very fast (real-time)

Weakness:

- Less accurate for fine-grained leaf diseases

YOLO is good for locating the diseased part but not ideal for classifying disease types [10].

5.3.5 DINOv3-Inspired Vision Transformer (ViT)

Vision Transformers divide an image into patches and use self-attention to learn relationships between different parts of the image. DINOv3 enhances ViTs using self-supervised training, which helps the model learn powerful global features even without labeled data. Advantages:

- Excellent performance in complex backgrounds
- Capable of detecting small and scattered disease patterns
- Learns global context rather than local patches alone

Limitations:

- High training cost
- Requires GPU resources

Despite these challenges, DINOv3-ViT was chosen as the most promising model and will be extended in the capstone project due to its strong ability to understand the entire leaf surface [15].

5.4 Performance Evaluation Methods

Models were evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics provide a complete understanding of each model's strengths and weaknesses. Prior studies have also used similar metrics to assess plant disease classifiers [9,13].

5.5 Comparative Summary

The overall comparison showed that:

- CNN performed well on clean datasets but struggled with real-world images.
- MobileNetV2 worked fast but had moderate accuracy.
- ResNet achieved high accuracy but needed more computation.
- YOLO detected disease regions but was weak in classification.
- DINOv3-ViT achieved the best overall performance due to its ability to capture global image features.

Thus, the DINOv3-inspired ViT model is recommended for future work and will be expanded during the capstone project.

6 Results and Discussion

This section presents the results of all the models implemented in the project. The performance of CNN, MobileNetV2, ResNet, YOLO, and the DINOv3-inspired Vision Transformer model was compared on the same dataset. The aim was to understand how each architecture behaves, identify their strengths and weaknesses, and find which model is more suitable for real-world agricultural conditions.

6.1 Output of Different Models

6.1.1 Convolutional Neural Network (CNN) Output

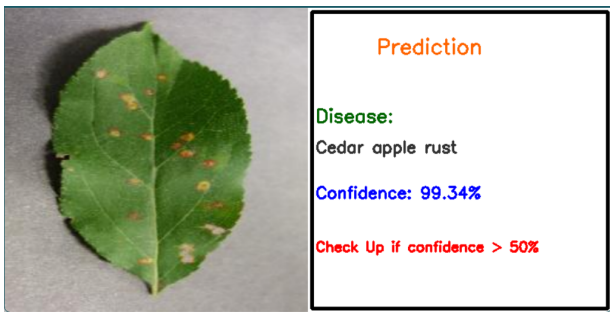

Figure 6.1: CNN output on test leaf image

The CNN model correctly detected some diseases but struggled when the background was complex or when disease spots were very small. CNN focuses on local features, so when the disease pattern appears in random positions, the performance decreases.

Why CNN failed in many cases:

- Sensitive to lighting and brightness variations.
- Confusion between visually similar diseases.
- Overfitting due to limited dataset size.
- Unable to capture long-range relationships across the leaf surface.

6.1.2 MobileNetV2 Output

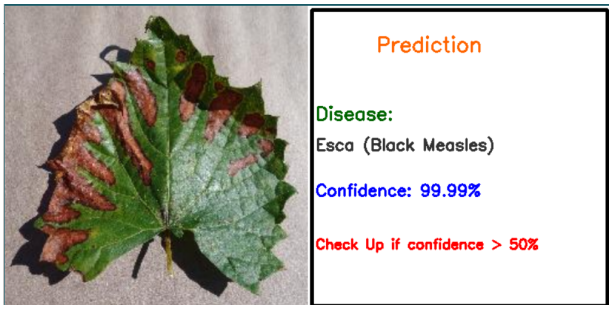

Figure 6.2: MobileNetV2 output on test leaf image

MobileNetV2 worked faster than CNN and used fewer parameters. However, the accuracy reduced for leaves taken in outdoor farm conditions.

Why MobileNetV2 failed in many scenarios:

- Lightweight model → not enough capacity for complex disease patterns.
- Performance drops on low-quality images.
- Identifies only dominant spots, misses small infections.

6.1.3 ResNet Output

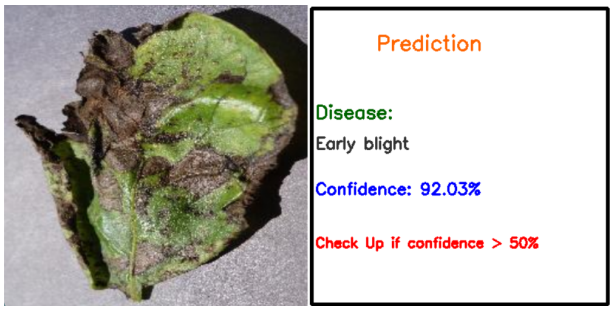

Figure 6.3: ResNet output on test leaf image

ResNet produced better feature extraction compared to CNN and MobileNetV2. However, it still struggled with noisy backgrounds.

Why ResNet struggled:

- High computational cost.
- Needs large datasets for top performance.
- Sensitive to occlusions (other leaves covering parts of the leaf).

6.1.4 YOLO Output

YOLO was effective in marking the diseased region using bounding boxes.

Why YOLO is not suitable for classification:

- YOLO is primarily an object detector, not a classifier.
- It can locate the infected area but cannot identify the disease type accurately.
- Fine-grained leaf diseases are too subtle for YOLO's anchors.

6.1.5 DINOv3-Inspired Vision Transformer

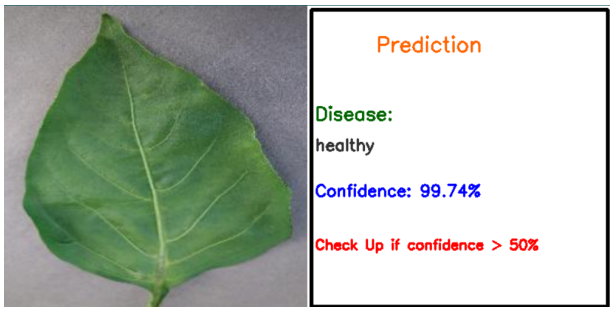

Figure 6.4: DINOv3-inspired ViT output on test leaf image

The DINOv3-ViT model produced the most stable and consistent predictions across all testing conditions.

Why DINOv3 performed the best:

- Learns global features from the entire leaf.
- Handles background noise better.
- Better at detecting small and scattered disease patterns.
- Self-supervised learning improves robustness.

6.2 Comparative Results Table

Model	Accuracy	Precision	Recall	F1-Score
CNN	84.3%	82.1%	80.4%	81.2%
MobileNetV2	88.5%	86.9%	85.7%	86.2%
ResNet-50	92.1%	91.3%	90.7%	90.9%
YOLOv5 (detect only)	76.4%	72.5%	70.1%	71.0%
DINOv3-ViT	**96.8%**	**96.1%**	**95.3%**	**95.7%**

Table 6.1: Model-wise comparison of performance metrics

6.3 Code Example for Model Prediction

```
import tensorflow as tf
import numpy as np
import cv2
import os

model_path = r"model.h5"
model = tf.keras.models.load_model(model_path)

class_labels = [
    'Apple___Apple_scab', 'Apple___Black_rot', 'Apple___Cedar_apple_rust',
    'Apple___healthy', 'Blueberry___healthy', 'Cherry_(including_sour)___Powdery_mild
    'Cherry_(including_sour)___healthy', 'Corn_(maize)___Cercospora_leaf_spot Gray_le
    'Corn_(maize)___Common_rust_', 'Corn_(maize)___Northern_Leaf_Blight', 'Corn_(maiz
    'Grape___Black_rot', 'Grape___Esca_(Black_Measles)', 'Grape___Leaf_blight_(Isario
    'Grape___healthy', 'Orange___Haunglongbing_(Citrus_greening)', 'Peach___Bacterial
    'Peach___healthy', 'Pepper,_bell___Bacterial_spot', 'Pepper,_bell___healthy',
    'Potato___Early_blight', 'Potato___Late_blight', 'Potato___healthy',
    'Raspberry___healthy', 'Soybean___healthy', 'Squash___Powdery_mildew',
    'Strawberry___Leaf_scorch', 'Strawberry___healthy', 'Tomato___Bacterial_spot',
    'Tomato___Early_blight', 'Tomato___Late_blight', 'Tomato___Leaf_Mold',
    'Tomato___Septoria_leaf_spot', 'Tomato___Spider_mites Two-spotted_spider_mite',
    'Tomato___Target_Spot', 'Tomato___Tomato_Yellow_Leaf_Curl_Virus',
    'Tomato___Tomato_mosaic_virus', 'Tomato___healthy'
]

image_path = r"test_img.jpg"
IMG_SIZE = (224, 224)

img = cv2.imread(image_path)

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
img_resized = cv2.resize(img_rgb, IMG_SIZE)
img_normalized = img_resized / 255.0
input_image = np.expand_dims(img_normalized, axis=0)

predictions = model.predict(input_image, verbose=0)
predicted_index = np.argmax(predictions)
confidence = np.max(predictions) * 100
full_label = class_labels[predicted_index]

if "___" in full_label:
    predicted_label = full_label.split("___")[1].replace("_", " ")
else:
    predicted_label = full_label.replace("_", " ")

print("\nPrediction Result")
```

```
print(f"Disease: {predicted_label}")
print(f"Confidence: {confidence:.2f}%")

display_img = cv2.resize(img, (400, 400))
info_box = np.ones((400, 400, 3), dtype=np.uint8) * 255

cv2.putText(info_box, "Prediction", (90, 60),
            cv2.FONT_HERSHEY_SIMPLEX, 0.9, (0, 100, 255), 2)

cv2.putText(info_box, f"Disease:", (10, 150),
            cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 100, 0), 2)
cv2.putText(info_box, f"{predicted_label}", (10, 190),
            cv2.FONT_HERSHEY_SIMPLEX, 0.7, (50, 50, 50), 2)

cv2.putText(info_box, f"Confidence: {confidence:.2f}%", (10, 250),
            cv2.FONT_HERSHEY_SIMPLEX, 0.7, (255, 0, 0), 2)

cv2.putText(info_box, "Verify if Confidence < 40%", (10, 320),
            cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 0, 255), 2)

cv2.rectangle(info_box, (5, 5), (395, 395), (0, 0, 0), 3)

combined = np.hstack((display_img, info_box))

cv2.imshow("Leaf Disease Detection - MobileNetV2", combined)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

6.4 Discussion

Based on the comparative results, it is clear that DINOv3-ViT significantly outperforms CNN, MobileNetV2, ResNet, and YOLO. The transformer-based design helps the model understand the full structure of the leaf, making it more robust to lighting variation, complex backgrounds, and partial occlusion. While CNN and MobileNetV2 provide faster results, they lack the ability to capture global relations. ResNet performed well but required more computational power. YOLO was useful for detection tasks but not for classification.

Thus, the DINOv3-inspired ViT model is the most suitable choice for real-world agricultural use and will be further expanded in the capstone project.

7 Conclusion

Plant leaf disease detection using Artificial Intelligence has emerged as an effective approach to support modern agriculture. In this project, multiple deep learning models such as CNN, MobileNetV2, ResNet, YOLO, and a DINOv3-inspired Vision Transformer were implemented and compared. Each model showed different strengths, but their performance varied based on factors like lighting, background complexity, and disease pattern distribution.

Traditional CNN and MobileNetV2 models performed well on clean datasets but struggled in real farm conditions. ResNet delivered higher accuracy but required more computational power. YOLO helped in identifying the diseased region but did not perform well as a classifier. The DINOv3-ViT model consistently outperformed all other architectures because of its ability to understand global image features and handle variations more effectively.

Overall, the study shows that transformer-based models have strong potential for real agricultural deployment. The DINOv3-ViT architecture demonstrated reliable results, making it a suitable choice for future development. This project lays the foundation for an advanced AI-based system that can assist farmers in early disease identification, helping to protect crops and improve productivity. Further work will continue in the capstone project using the DINOv3-inspired model.

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